

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

2. Let $G = (V, E)$ be the graph with vertices $V = \{0, 1, 2, 3, 4, 5\}$ and edges $E = \{(0, 2), (0, 5), (1, 2), (1, 5), (0, 1), (2, 3), (0, 4), (1, 4), (3, 5), (2, 5)\}$.

- (a) Compute the breadth first search tree for G and for each vertex compute its degree. Show all steps.
- (b) What is the asymptotic cost of the vertex degree computation relative to the construction of the tree?

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Answer:

2. (a) The construction of the breadth-first tree layers the vertices.
- i. 2, 5, 1, 4 are all vertices connected to vertex 0, with edges $\{(0, 2), (0, 5), (0, 1), (0, 4)\}$.
The degrees are $\deg(0) = 4$, $\deg(2) = 4$, $\deg(5) = 4$, $\deg(1) = 4$, and $\deg(4) = 2$.
 - ii. 3 is the only vertex left, connected to 2; the edges are $\{(0, 2), (0, 5), (0, 1), (0, 4), (2, 3)\}$.
The degree: $\deg(3) = 2$.
- (b) For a graph of n vertices and m edges, the breadth-first search algorithm runs in time $O(n + m)$.
- To compute the degrees of each vertex, we have to compute the length of all edge lists which requires $2m$ steps, which is $O(m)$.
- As $O(n + m) + O(m)$ is $O(n + m)$, the vertex computation does not asymptotic cost of the breadth-first search algorithm.